O-I-554B
March 24, 1977
SUPERSEDING
O-I-554A
March 1, 1973

FEDERAL SPECIFICATION

INSECTICIDE, MALATHION, POWDER, DUSTING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one type of insecticide, malathion, powder, dusting (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

RR-S-366	-	Sieve, Test
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-D-723	-	Drums, Fiber
PPP-P-704	-	Pails, Metal: (Shipping, Steel, 1 Through 12 Gallon)

Federal Standards:

FED-STD-123	-	Marking for Shipment (Civil Agencies)
FED-STD-141	-	Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-B-26701	-	Bottles, Screw Cap and Carboys, Polyethylene, Plastic
MIL-M-51064	-	Insecticide, Malathion

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized and Containerized Unit Loads 40 Inch X 48 Inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-1233	-	Procedures for Determining Particle Size, Particle Size Distribution, and Packed Density of Powdered Materials

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Federal Insecticide, Fungicide and Rodenticide Act, as amended (7 USC, Section 136 et seq.)

(Application for copies should be addressed to the Environmental Protection Agency, Pesticide Registration Division, 401 M Street, S. W., Washington, DC 20460.)

Title 49, Code of Federal Regulations (CFR) 170-189, Department of Transportation Rules and Regulations for Transportation of Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

GCMI Recommended Glass Finish 410

(Application for copies should be addressed to the Glass Container Manufacturers Institute, 1800 K Street, N.W., Washington, DC 20006.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Composition and form. The finished insecticide shall be a fine free-flowing powder made by thoroughly mixing a pyrophyllite diluent with premium grade malathion. The insecticide shall contain a minimum of 1.00 or a maximum of 1.10 percent malathion when determined as specified in 4.3.1.

3.2 Malathion. The malathion shall conform to the requirements of MIL-M-51064, grade A.

3.2.1 Diluent. The diluent shall be pyrophyllite only and of such degree of fineness that not less than 99 percent will pass through a U.S. Standard No. 100 (149 microns) sieve, when tested as specified in 4.3.3.

3.3 Moisture. The finished insecticide shall not lose more than 2 percent of its weight when tested as specified in 4.3.2.

3.4 Fineness. The finished insecticide powder shall be a fine, free-flowing material devoid of lumps. The powder shall be of such fineness that not less than 99 percent will pass through a U.S. Standard No. 100 (149 microns) sieve, when tested as specified in 4.3.3.

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3.5 Surface mean particle diameter. The finished insecticide powder shall have a surface mean particle diameter of not more than 5 microns when determined as specified in 4.3.4.

3.6 pH. A suspension of the finished insecticide powder shall have a pH value not lower than 5.0 and not higher than 7.0 when tested as specified in 4.3.5.

3.7 Labeling. Labeling shall be in compliance with the requirements of the Federal Insecticide, Fungicide, and Rodenticide Act, as amended. The labels shall be approved by and registered with the Pesticide Registration Division of the Environmental Protection Agency (EPA).

3.7.1 Label data. The label data shall be durably and legibly marked on the main panel of the body by lithographing, stenciling, or printing, and silk screening. Yellow ink shall be used on pails and black ink shall be used on polyethylene bottles. If the label data are too voluminous to be applied in this manner, instructions for use may be placed on a label made of white paper as specified in MIL-STD-129. Labels shall be securely affixed in place with water resistant label adhesive as specified in MIL-STD-129 and shall be waterproofed by coating the entire outer surface of the label with the same adhesive.

3.7.1.1 Three ounce (85.0g) polyethylene bottles containing 2-ounce (56.7 g) of malathion powder. The label for the bottles shall include, but not be limited to, the following information:

EPA REGISTRATION NUMBER
EPA ESTABLISHMENT NUMBER

INSECTICIDE, MALATHION, POWDER, DUSTING
FEDERAL SPECIFICATION O-I-554
NATIONAL STOCK NUMBER
NET CONTENTS - 2 oz (56.7 g)
MANUFACTURER'S NAME AND ADDRESS
CONTRACT AND LOT NUMBER
DATE OF MANUFACTURE

CAUTION

KEEP OUT OF REACH OF CHILDREN

For external use only. Keep away from food and mess gear. Do not use if insect repellent or other oily liquids are applied to skin. Do not apply more than two ounces (56.7 g) within seven full days. Do not use for at least seven days after you have been dusted with a hand or machine duster. Avoid breathing dust. Avoid contact with eyes. In case of contact immediately flush eyes with plenty of water. Get medical attention if irritation persists. Wash hands thorougly after using. If a skin reaction appears discontinue use and consult a physician.

Note to physicians: Malathion may cause cholinesterase inhibition. Atropine is antidotal.

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Active ingredient:	Percent by weight
Malathion (O,O-dimethyl phosphorodithioate of diethylmercaptosuccinate)	1 percent
Inert ingredient:	
Pyrophyllite	99 percent

DIRECTIONS FOR USE

FOR BODY LICE: If possible, bathe before using. Dust entire inner surfaces of undergarments, especially seams; also inner seams of trousers and shirt. Rub treated areas lightly by hand to spread powder evenly. Treatment will require about 1 ounce (28.4 g) or one-half the contents of this container. If clothing cannot conveniently be removed, unbutton the shirt and trousers and dust the powder liberally by shaking it on the inside of underwear or other garment next to the skin. Then pat the clothes with the hands to insure even distribution. Repeat at weekly intervals or following each change of clothing until infestation is controlled.

FOR HEAD LICE: Apply to head rubbing the hair lightly with fingertips until the hair is whitened. Do not wash the hair for at least 24 hours. Repeat at the end of 1 and 2 weeks.

FOR PUBIC (CRAB) LICE: Apply to all regions of body where coarse hair is present. Thorough application is important. Do not bathe for at least 24 hours. Repeat at end of 1 and 2 weeks. The use of this product for other than treatment of body, head or public lice is a violation of Federal law.

3.7.1.2 Twenty-five pound (11.34 kg) containers. The label for the container shall include, but not be limited to, the following information:

EPA REGISTRATION NUMBER
EPA ESTABLISHMENT NUMBER

INSECTICIDE, MALATHION, POWDER, DUSTING
FEDERAL SPECIFICATION O-I-554
NATIONAL STOCK NUMBER
NET CONTENTS - 25 lbs (11.34 kg)
MANUFACTURER'S NAME AND ADDRESS
CONTRACT AND LOT NUMBER
DATE OF MANUFACTURE

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CAUTION

KEEP OUT OF REACH OF CHILDREN

For external use only. Avoid breathing dust. Use with adequate ventilation. Avoid contact with eyes. In case of contact, immediately flush eyes with plenty of water. Get medical attention if irritation persists. Keep away from food or mess gear. This product is toxic to fish. Keep out of lakes, streams or ponds. Do not contaminate water by cleaning of equipment or disposal of wastes. Apply this product only as specified on this label.

Personnel engaged in mass delousing using this material shall wear respirators approved for use by the National Institute of Occupational Safety and Health (NIOSH). Unlined rubber gloves should be used for handling, and prolonged skin contact with large amounts of the dust shall be avoided. Shower at the end of the day. Uniforms shall be laundered before wearing again.

Note to physicians. Malathion may cause cholinesterase inhibition. Atropine is antidotal.

Active ingredient:	Percent by weight
Malathion (O,O-dimethyl phosphorodithioate of diethylmercaptosuccinate)	1 percent
Inert ingredient:	
Pyrophyllite	99 percent

STORAGE INSTRUCTIONS

Store in a cool, dry place when possible. Never exceed 90°F storage temperature. The average effective period of this item is 12 months from date of manufacture. If used after this period, malathion content should not be below 0.75 percent.

DIRECTIONS FOR USE

1. When dusting by hand or machine, do not apply more than two ounces (56.7 g) to each person.

2. Do not redust for seven (7) full days after a previous dusting.

3. Individual 3 ounce (85.0 g) polyethylene bottles containing 2 oz (56.7 g) of Malathion powder will not be used by personnel who have been dusted by hand or machine for at least seven (7) full days after such dusting

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4. If possible, troops should bathe before being dusted.

5. Do not dust individuals who have insect repellents, oil, or grease on the skin.

6. The use of this product for other than treatment of body, head, or pubic lice is a violation of Federal law.

INSTRUCTIONS FOR USE

FOR BODY LICE:

1. Clothing off person: Turn clothing inside out and dust lightly over entire inner surface, rubbing powder well into all seams. Repeat at weekly intervals until infestation is controlled.

2. Clothing on person: Use machine or hand duster. If hand duster, fill 3/4 full and adjust to deliver heavy cloud of powder at each stroke. One hand duster 3/4 full holds enough powder to treat approximately 10 individuals. With either the machine or hand duster the standard 16-point delousing technique should be used. For each point, deliver a 3-second burst (machine) or 2 full strokes (hand).

a. Dust the subject's head, having him rub the powder into his hair until whitened, then dust inside hat and replace on head (points 1 and 2).

b. Insert nozzle into right sleeve next to skin. With subject's arm outstretched to the side at shoulder height, direct powder toward armpit. Repeat for left sleeve (points 3 and 4). The subject's face should be turned away from the side being dusted.

c. Insert nozzle in the front of the shirt at collar and direct the powder toward the right armpit (point 5), toward the stomach (point 6) and toward the left armpit (point 7). Operator stands in front, and subject leans forward with head tipped back.

d. Insert nozzle in front of trousers, next to skin, and direct powder toward right leg (point 8), toward pubic region (point 9) and toward left leg (point 10).

e. Insert nozzle in back of shirt at collar and direct powder toward right shoulder (point 11), toward small of back (point 12), and toward left shoulder (point 13). Operator and subject remain in same relative position as above but with head of subject bent on chest.

f. Insert nozzle in back of trousers, next to skin, and direct powder toward right leg (point 14), toward buttocks crease (point 15) and toward left leg (point 16).

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FOR HEAD AND PUBIC LICE: Dust into infested hairy portions of the body and work down to the base of the hair with finger tips. Repeat treatment at end of 1 and 2 weeks.

3.8 Workmanship. The finished product shall be clean, uniform, devoid of lumps and free of defects which might impair its utility.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase documents.

4.2.1.1 Testing of components. Tests shall be performed on samples from each lot or batch of components and materials used in the composition of the end item for the applicable characteristics listed in table I. The sample unit for each component shall be 1/2 pound (227 g). The sample size shall be five. The test sample for each component shall be a pound (454 g) composite of all sample units. There shall be no failure to meet the specified requirements, or duplicate determinations on the composite.

4.2.1.2 Composite sample. Sample with a trier long enough to reach the bottom of the container at one edge or corner and probe diagonally toward opposite edge or corner. Use airtight glass jars of at least one-pound (454 g) capacity for sample storage. After thorough mixing of all sample units, a one-pound (454 g) composite sample shall be drawn for testing purposes (see table I).

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4.2.2 Inspection of the end item.

4.2.2.1 Examination of the end item. The end item shall be examined in accordance with the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.2.2.1.6. Random samples shall be drawn from each lot of the end items for examination of visual and preparation for delivery defects. The lot shall consist of all material submitted for inspection at one time. The lot, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled primary containers for examination in 4.2.2.1.1, 4.2.2.1.2, 4.2.2.1.3 and in units of shipping containers for examination in 4.2.2.1.4, and in units of pallets for the examination in 4.2.2.1.5.

4.2.2.1.1 Examination of the end item container for defects in closure, marking appearance, construction, and workmanship. The sample unit for this examination shall be one filled container.

Examine	Defect
Closure	Any leakage or sifting of contents.
Appearance, construction and workmanship	Dented, cracked, or crushed affecting serviceability. Indentation of rust (if applicable). Not in container specified. Sifter, as applicable, not of type or size specified. Not clean - any evidence of dirt, dust or foreign matter. Not specified finish. Enamel not thoroughly dry (tacky).
Markings (as applicable)	Omitted; illegible; incorrect, incomplete, not in accordance with requirements. Label data not lithographed (as required) with black ink; not clear, smears.

4.2.2.1.2 Examination of the end item container for defects in net contents. The sample unit for this examination shall be one filled container.

Examine	Defect
Net contents	Average contents less than specified or indicated quantity.

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4.2.2.1.3 Examination of the end item container for defects in appearance of insecticide. The sample unit for this examination shall be one filled container.

Examine	Defect
Appearance of insecticide	Presence of foreign matter. Not homogeneous. Not in powder form. Contains lumps. Not free flowing.

4.2.2.1.4 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: drum, fiberboard container, or partitions not as specified; incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container.

4.2.2.1.5 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

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4.2.2.1.6 Inspection level and acceptable quality level (AQL). The inspection levels and AQLs, expressed in defects per hundred units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	I	1.5
4.2.2.1.2	S-2	-
4.2.2.1.3	S-2	4.0
4.2.2.1.4	S-2	4.0
4.2.2.1.5	S-1	6.5

4.2.3 Testing of the end item. The end item shall be tested for the applicable characteristics as specified in table I for each lot presented for acceptance. The sample unit shall be one pound (454 g) of insecticide. The sample size shall be five. Each sample unit shall be taken from a different shipping container of the lot. The test sample shall consist of a two pound (907 g) composite of the five sample units. There shall be no failure to meet the requirements for each characteristic on duplicate determinations on the composite sample. Compliance with the requirement in 3.1 that the insecticide dusting powder shall contain only the material specified shall be satisfied by acceptance of a certificate of compliance from the supplier.

4.3 Test procedure.

4.3.1 Malathion content. Malathion, O,O-dimethyl phosphorodithioate of diethylmercaptosuccinate, is decomposed by alkali in ethanol to sodium O,O-dimethyl dithiophosphate, sodium fumarate, and ethanol. The sodium O,O-dithiophosphate is then converted to the cupric complex which is soluble in cyclohexane with the formation of an intense yellow color. The color intensity is proportional to the concentration of O,O-dimethyl dithiophosphoric acid and is measured colorimetrically at 420 millimicrons, the absorption peak. The corresponding amount of insecticide is derived by comparison with a standard prepared with a known amount of pure insecticide, which is simultaneously carried through the procedure.

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TABLE I. Testing of components and finished insecticide

Component and characteristic	Specification reference		Number of determinations on composite sample 1/	Results reported to nearest
	Requirement	Test method		
Composition	3.1	4.3.1	2	0.02 percent
Pyrophyllite, fineness (dry)	3.2.1	4.3.3	2	0.5 percent
Finished insecticide:				
Moisture	3.3	4.3.2	2	0.1 percent
Fineness (dry)	3.4	4.3.3	2	0.5 percent
Particle diameter	3.5	4.3.4	2	0.1 micron
pH	3.6	4.3.5	2	0.1 percent

1/ Test reports shall include all values on which results are based.

In this method, ferric reagent is added to oxidize materials which would reduce cupric ions to cuprous ions; with dithiophosphoric acid, cuprous ions form a colorless complex which is apparently more stable than the yellow cupric complex.

4.3.1.1 Reagents.

(a) Malathion reference standard (see 6.6).
(b) Acetonitrile. Boiling range, 80°-82°C, neutral to litmus.
(c) Sodium hydroxide solution. 0.5N aqueous solution: Dissolve 4 grams (g) of sodium hydroxide in 200 milliliters (ml) of water.
(d) Ferric reagent. Dissolve 4.0 g of ferric chloride ($FeCl_3 \cdot 6H_2O$) plus 160 ml of hydrochloric acid in water and dilute to 500 ml. Dilute 25 ml of this solution with water to 1000 ml.
(e) Copper sulfate solution. Dissolve 1.5 g of copper sulfate ($CuSO_4 \cdot 5H_2O$) in 100 ml of water.

4.3.1.1.1 Apparatus. Spectrophotometer or photoelectric colorimeter with a blue filter (420 millimicrons) and one-centimeter absorption cells.

4.3.1.2 Preparation of standards.

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4.3.1.2.1 Caution. It is very important in preparing the standard and specimen solutions that volume fluctuations with changing temperatures be avoided. This is accomplished by bringing each solution to the volume mark just before proceeding to the next step in the procedure, as indicated.

4.3.1.2.2 Procedure. Weigh 1.000 ± 0.020 g of malathion reference standard into a 5 or 10 ml beaker. Quantitatively transfer to a 1000 ml volumetric flask with anhydrous ethanol. Dilute with anhydrous ethanol to approximately 5 ml below the volume mark. Allow the solution to come to a selected temperature. Record the temperature. Dilute the solution with anhydrous ethanol to the volume mark and mix well. Call this the "Standard Solution A." (Note: Standard Solution A may be retained for daily use for a period not to exceed two weeks. It is very important that it be adjusted to the temperature recorded above before aliquots are withdrawn.) Using a volumetric pipet, transfer a 15 ml aliquot of Standard Solution A to a 250 ml volumetric flask. Add 2.5 ml of acetonitrile and dilute with anhydrous ethanol to approximately 5 ml below the volume mark. Call this "Standard for Analysis". Retain this solution until Specimen for Analysis is prepared and ready for final dilution. At that time, dilute with anhydrous ethanol to the volume mark and mix well.

4.3.1.3 Preparation of specimen. Weigh accurately a specimen equivalent to 0.300 ± 0.020 g of real malathion (approximately 30 g of 1 percent dust) and quantitatively transfer to a 4 ounce (113 g) bottle. Using a volumetric pipet, add 50 ml of acetonitrile. Cap the bottle tightly and shake vigorously for 2 or 3 minutes either manually or on a reciprocating shaker. Allow the solids to settle for 3 to 5 minutes. If necessary, centrifuge for 2 to 3 minutes at 1500 to 2000 rpm. Call the supernatant liquid "Specimen Solution A". Immediately, using a volumetric pipet, transfer a 25 ml aliquot of Specimen Solution A to a 250 ml volumetric flask. Dilute with anhydrous ethanol to the volume mark and mix well. Call this "Specimen Solution B". Immediately, using a volumetric pipet, transfer a 25 ml aliquot of Specimen Solution B to a 250 ml volumetric flask. Dilute with anhydrous ethanol to the volume mark and mix well. Call this "Specimen for Analysis".

4.3.1.4 Analysis. To obtain accurate results, it is important that the Standard for Analysis be carried along with each Specimen for Analysis through the following procedure without interruption:

Using a volumetric pipet, transfer 25 ml aliquots of Standard for Analysis and Specimen for Analysis to separate 250 ml separatory funnels. Add 2 ± 0.1 ml of 0.5N aqueous sodium hydroxide to each sample and mix well by swirling gently (do not shake) for 5 to 10 seconds. Allow the solution to stand for a total elapsed time of 2 minutes ± 10 seconds. During this waiting period, measure two 75 ± 1 ml portions of ferric reagent. At the end of the period, add the ferric reagent to each separatory funnel, swirl for 10 seconds. Allow the solution to stand for a total elapsed time of 5 minutes. During this waiting period, fill two 50 ml volumetric flasks to the volume mark with cyclohexane. Simultaneously transfer the cyclohexane from each volumetric flask to the respective separatory funnel, allowing the flasks to drain one minute.

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From this point on, the Specimen for Analysis and Standard for Analysis must be handled separately. Carry each through to measurement of absorbance without interruption. The next step is very critical and should be carried out exactly.

Add 2.0 ± 0.1 ml of copper reagent. Start shaking the separatory funnel immediately after the addition of copper reagent and shake it for exactly 1 minute. (Note: The malathion-copper complex is unstable in the aqueous phase. The copper reagent should be added with a fast delivery Mohr pipet and the shaking started with practically no time elapse. A delay of only 15 seconds between the addition of copper-reagent and the shaking will result in an absorbance reading which is as much as 4 percent lower than when shaking is started immediately. Longer delays result in much lower absorbance readings with losses up to 40 percent resulting after a 2-minute delay.) Allow the phases to separate. As soon as separation occurs, discard the aqueous phase. Allow some solvent to rinse the stem of the separatory funnel, then run the clear solvent phase into a small beaker for transfer to an absorption cell or run the clear solvent phase directly into the cell. (Note: Manipulate the separation of the phases of both standard and specimen solutions in a similar manner.)

Measure the absorbance of the yellow solution within 5 minutes at 420 millimicrons, using cyclohexane as the reference. (Note: Absorbance measurements of both standard and specimen solutions are made at the same fixed time after the end of the shaking period. The yellow color of the copper complex in cyclohexane is unstable, so the color should be read within 5 minutes after extraction. During this time the absorbance reading drops very slowly.)

4.3.1.5 Calculation. The absorbance measurements of the standard and specimen solutions should agree within + 10 percent of each other.

$$\text{Percent malathion} = \frac{A\ sp}{A\ std} \times \frac{\text{grams of standard}}{\text{grams of specimen}} \times \text{purity of standard} \times 0.3$$

Where: A sp = the absorbance of Specimen for Analysis at 420 millimicrons, and
A std = the absorbance of Standard for Analysis at 420 millimicrons

4.3.2 Moisture. Weigh 5 grams of the insecticide powder, dusting into a tared moisture dish (nickel, platinum, or aluminum) and dry for 10 hours under atmospheric pressure at temperature of 100°C (± 2°C). Cool in desiccator and weigh. Dry again for 1 hour or until change in weight is not more than 2 milligrams. Calculate the percent loss and report it as moisture.

4.3.2.1 Calculation.

$$\text{Percent loss} = \frac{\text{loss of weight}}{\text{weight of sample}} \times 100$$

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4.3.3 Fineness. Screen a dry 20 gram sample through a U.S. Standard sieve of the designated mesh size conforming to RR-S-366 using a single eccentric type mechanical shaker which imparts to the sieve a rotary motion and tapping action of uniform speed of 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Continue the screening for 15 minutes. Weigh the residue and calculate the percentage passing through the sieve.

4.3.4 Surface mean particle diameter. The surface mean particle diameter shall be determined by the air permeation method. The apparatus used shall be a Government approved air-permeation apparatus properly calibrated (see 6.3). The solid density of the powder, knowledge of which is prerequisite to particle fineness measurement by the air-permeation procedures, shall be determined in accordance with method 4181 of FED-STD-141, with the following exceptions:

a. The immersion medium shall be a one percent solution of a suitable wetting agent in distilled water (see 6.4).
b. Omit the oven drying of the sample.

4.3.5 pH. Shake a suspension of 20 grams of the finished insecticide and 80 milliliters of distilled water in a glass-stoppered Erlenmeyer flask at intermittent intervals, during the course of one hour. Filter the suspension through a quantitative filter paper and determine the pH of the filtrate by any suitable method. In the event of a dispute, however, the results obtained with a calibrated pH electrometer shall govern.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A. The insecticide shall be packaged in 3 ounce (85 g) polyethylene bottles, and 25 pound (11.34 kg) containers as specified (see 6.2).

5.1.1.1 Unit package.

5.1.1.1.1 Three ounce polyethylene bottle. Two ounces (56.7 g) of the insecticide shall be unit packaged in a 3 ounce (85 g) polyethylene bottle conforming to MIL-B-26701 with the following exceptions:

The bottle shall be oval around the vertical axis, and shall be formed in one piece with a 20-410 medium (depth) continuous thread, concealed bead finish in accordance with the standard of the Glass Container Manufacturers Institute. The approximate dimensions of the bottle shall be 4 3/32 inches (104 mm) overall height, with a major axis of 2 1/4 inches (57 mm) and a minor axis of 1 1/8 (29 mm) inch. The minimum wall thickness shall be 0.015

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inch (0.4 mm). The weight of each empty bottle less cap and plug shall be 15 grams ± 1/2 gram. The inside dimensions of the finish shall be 0.544 to 0.558 inches (13.8 to 14.2 mm). Each finish shall be provided with an 0.565 ± 0.005 inch (14.4 ± 0.1 mm) diameter tube well, vertical spray orifice type plug. Each plug shall have an 0.037 ± 0.003 inch (0.9 ± 0.1 mm) diameter spray orifice. Each plug shall be furnished with a polyethylene tube having an inside diameter of 0.052 ± 0.003 inch (1.3 ± 0.1 mm) and an outside diameter such that the tube will fit snugly into the tube well of the plug. Each bottle shall be provided with a cap made of black phenolic or melamine thermosetting plastic. The cap shall match the 20-410 finish of the bottle. The cap shall be furnished with a securely fitted 0.040 inch (1.0 mm) thick cork cap liner faced with 0.0015 inch (0.04 mm) thick tin or aluminum foil. When the cap is completely screwed onto the bottle, the plug shall fit firmly against the cap insert to assure sealing against dusting. There shall be not less than 1/32 inch (0.8 mm) clearance between the bottom skirt of the cap and any portion of the shoulder area of the bottle. Each bottle shall be sealed in accordance with the applicable secondary closure requirements of MIL-B-26701.

5.1.1.1.2 Twenty-five pound containers. Twenty-five pounds (11.34 kg) of insecticide shall be unit packaged in a 25 pound (11.34 kg) container as hereinafter specified.

5.1.1.1.2.1 Metal pail. The metal pail shall conform to type II, class 4 of PPP-P-704 and shall be finished with an interior coating (see 6.5). Each pail shall be closed in accordance with the applicable closure requirements specified in the appendix of PPP-P-704. There shall be a metal tag securely wired to a lug with the Federal Identification Stock No. embossed on it.

5.1.1.1.2.2 Fiber drum. The fiber drum shall conform to type III, grade A, class 2 of PPP-D-723. Each drum shall be fitted with a 2 ply bag. The bag's inner ply shall be kraft paper coated with polyethylene or laminated with glassine. Each drum shall be closed as specified in PPP-D-723.

5.1.1.2 Intermediate package.

5.1.1.2.1 Three ounce (85.0g) polyethylene bottles. Forty-eight 3 ounce (85.0g) polyethylene bottles of insecticide, unit intermediate packaged as specified in 5.1.1.1, shall be packaged in a snug-fitting fiberboard box conforming to style RSC, grade W6c of PPP-B-636. Each box shall be provided with full height, half-slotted style interlocking partitions and a tier pad (when applicable) fabricated of the same material as the box. Each box shall be closed in accordance with method IV as specified in the appendix of the box specification.

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5.1.1.2.2 Twenty-five pound (11.34 kg) containers. Twenty-five pound (11.34 kg) containers of insecticide, unit packaged as specified in 5.1.1.1, shall not require intermediate packaging.

5.1.2 Level C. Insecticide shall be packaged to afford adequate protection against physical damage and deterioration during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Three ounce (85.0g) polyethylene bottles. One hundred and ninety-two 3 ounce (85.0g) polyethylene bottles of insecticide, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.2.1.4.

5.2.1.2 Twenty-five pound (11.34 kg) containers. Insecticide, packaged in 25 pound (11.34 kg) containers as specified in 5.1, shall not require overpacking.

5.2.2 Level B.

5.2.2.1 Three ounce (85.0g) polyethylene bottles. One hundred and ninety-two 3 ounce (85.0g) polyethylene bottles of insecticide, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.2.1.4.

5.2.2.1.1 Weather-resistant fiberboard containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.2.1.4.

5.2.2.2 Twenty-five pound (11.34 kg) containers. Insecticide, packaged in 25 pound (11.34 kg) containers as specified in 5.1, shall not require overpacking. Alternatively, the fiber drum specified in 5.1.1.1.2.2 may conform to type I, grade A, class 2 of PPP-D-723.

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5.2.3 Level C. Insecticide, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Palletization. When specified (see 6.2), insecticide, packed as specified in 5.2, shall be palletized in accordance with load type I or XIII (as applicable), of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Load type I pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of load type I loads shall be effected by reversing the pattern of each course. If the fiberboard container is of a size which does not conform to any of the load type I pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. Marking shall be in accordance with 5.4.1 or 5.4.2 as specified (see 6.2).

5.4.1 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129 and in accordance with the marking and labeling requirements of Regulations for Transportation of Explosives and Other Dangerous Articles.

5.4.2 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123 and in accordance with the marking and labeling requirements of Regulations for Transportation of Explosives and Other Dangerous Articles.

5.4.3 Additional marking for intermediate package and shipping containers. In addition to the marking above, intermediate packages and shipping containers shall be marked as follows:

STORAGE INSTRUCTIONS

Store in a cool, dry place when possible. Never exceed 90°F (32.2° C) storage temperature. The average effective period of this item is 12 months from date of manufacture. If used after this period, malathion content should not be below 0.75 percent.

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6. NOTES

6.1 Intended use. The insecticide, malathion, powder, dusting is intended for use in the control of body, pubic, and head lice.

6.2 Ordering data. Purchasers, should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size of package desired (see 5.1.1).
(c) Inspection responsibility, methods, and tests (see section 4).
(d) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing of 3 ounce (85.0g) polyethylene bottles (see 5.2.2.1.1).
(f) When palletization is required (see 5.3).
(g) Marking required (see 5.4).

6.3 Air-permeation apparatus. An approved apparatus is the self calculating air-permeation apparatus of the U.S. Department of Agriculture (Patent No. 2,261,802), or any apparatus conforming to MIL-STD-1233.

6.4 Wetting agent. A suitable wetting agent is Tergitol penetrant No. 4, manufactured by the Commercial Solvents Corporation, New York, NY.

6.5 Interior coating. Coatings proved satisfactory by long-term storage tests (see 5.1.1.1.2.1) are the following:

Inland Steel Company's IC-20, IC-26, IC-93, IC-98, and IC-101.
Continental Can Company's 224 Phenolic.
American Cyanamid's 2 coats of DB 124-17.
Interchemical Company's 2 coats of 117 Brown or 2 coats of 14-474 or 1 coat of 63A Prime and 1 coat of 117 Brown.

6.6 Reference standard. The malathion reference standard (see 4.3.1.1) may be obtained from:

Chem Service Incorporated
851 Lincoln Avenue
P.O. Box 194
West Chester, Pennsylvania 19380

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6.7 International interest. Certain provisions of this specification are subject of international standardization agreement as cited in ABCA QSTAG 232. When amendment, revision, or cancellation of this specification is proposed which affects or violates the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

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Army - GL

Review activities:

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Preparing activity:

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Civil Agency Coordinating Activities:

AGR - AMS, APHIS
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